Gauge-invariant slow modes in a seven-link SU(2) transfer: finite-coupling mixing and the failure of uniform fixed-step contraction

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Reviewed finite-model draft, October 29, 2026

Abstract

We study an explicitly normalized transfer on two adjacent plaquettes with seven SU(2) links. Independent central Wilson increments update each link once, and a spatial half weight is applied on each side of the kinetic operator. Positive kernel powers construct a uniformly positive bounded ground function at every fixed coupling, giving a genuine reversible stationary Markov kernel. For a sufficiently small positive spatial tilt, depending on the temporal coupling, a three-channel gauge-invariant trial has normalized stationary lag strictly larger than the bare one-plaquette character factor. The proof includes exact Haar projections, a positive supersolution, explicit cubic errors and a positive optimized second-order margin. Every fixed number of Markov steps fails to give a parameter-uniform subunit covariance factor, even after arbitrary random gauge actions. At each fixed coupling, a separate positive minorization gives geometric mixing. We also exhibit the loss of the trial's centering symmetry when both spatial plaquettes are weighted. All statements concern this finite model; no homogeneous four-dimensional transfer or physical mass-gap conclusion is inferred. Classical positive-kernel and character methods are credited; priority for the exact comparison is unresolved.

1 The model and the quantified conclusions

Identify $G = \mathrm{SU}(2)$ with unit quaternions, and write $\tau(g) = \frac{1}{2}\operatorname{Re}\operatorname{tr}(g)$ for its scalar quaternion coordinate. Let dg be normalized Haar measure. For $\beta \in \mathbb{R}$ define the actual probability law

$$Z_\beta = \int_G e^{\beta\tau(v)}\,dv, \qquad d\mu_\beta(v) = Z_\beta^{-1} e^{\beta\tau(v)}\,dv, \qquad c = c_\beta = \int_G \tau(v)\,d\mu_\beta(v). \tag{1}$$

The second scalar channel is

$$m_\beta = \int_G \tau(v)^2\,d\mu_\beta(v), \qquad r = r_\beta = \frac{4m_\beta - 1}{3}. \tag{2}$$

All these quantities are finite, and $Z_\beta > 0$, since $|\tau| \le 1$.

Our state space is $\mathcal{S} = G^7$, with coordinates $s = (U, A, B, C, D, E, F)$ and product Haar probability $dH(s)$. The six vertices are numbered $0, \dots, 5$. Orient the shared link U from 3 to 0, the path (A, B, C) from 0 through $1, 2$ to 3, and the path (D, E, F) from 3 through $4, 5$ to 0. Put

$$\begin{gathered} L = ABC, \quad R = DEF, \quad x(s) = \tau(LU), \quad y(s) = \tau(U^{-1}R), \\ z(s) = \frac{\tau(LR)}{4}, \qquad w(s) = x(s)y(s) - z(s). \end{gathered} \tag{3}$$

Thus x, y are the two adjacent plaquette traces; z is the shared-link Haar projection of their product. Every physical link occurs once in the noise law $\mu_\beta^{\otimes 7}$.

On bounded measurable real functions define

$$K_\beta f(s) = \int_{G^7} f(sv)\, d\mu_\beta^{\otimes 7}(v), \qquad h_t(s) = e^{tx(s)/2}, \qquad S_{\beta,t} = M_{h_t} K_\beta M_{h_t}. \tag{4}$$

Products sv are coordinatewise. The spatial parameter t and temporal parameter β are independent. Only the first plaquette is weighted in (4). Section 10 explains why that qualification cannot be removed from the slow-mode proof.

Theorem 1.1. *For each $\beta, t \in \mathbb{R}$, there are $\Lambda > 0$ and a bounded measurable $\psi : \mathcal{S} \to (0, \infty)$, bounded away from zero, such that $S_{\beta,t}\psi = \Lambda\psi$. They define a reversible Markov kernel*

$$P_{\beta,t} f(s) = \frac{S_{\beta,t}(\psi f)(s)}{\Lambda\psi(s)}, \qquad d\pi_{\beta,t} = \frac{\psi^2\, dH}{\int \psi^2\, dH}. \tag{5}$$

The ground function can be chosen even under the private-link center flip $D \mapsto -D$ and invariant under the six-vertex gauge group.

If $c_\beta \neq 0$, there is $\delta_\beta \in (0, 1]$ such that for every $0 < t < \delta_\beta$ a bounded gauge-invariant f satisfies

$$\int f\, d\pi_{\beta,t} = 0, \qquad \int f^2\, d\pi_{\beta,t} > 0, \qquad c_\beta^4 < \frac{\int f P_{\beta,t} f\, d\pi_{\beta,t}}{\int f^2\, d\pi_{\beta,t}}. \tag{6}$$

For the same f, every integer $n \geq 0$ obeys

$$1 - n(1 - c_\beta^4) \leq \frac{\int f P_{\beta,t}^n f\, d\pi_{\beta,t}}{\int f^2\, d\pi_{\beta,t}} \leq 1. \tag{7}$$

For each fixed n, $q < 1$ and $B \in \mathbb{R}$, one can choose $\beta > B$ and $0 < t \leq 1$ with a centered gauge-invariant f whose normalized n-step lag exceeds q. These conclusions persist if an arbitrary probability-distributed gauge action is applied before every Markov step.

This is a statement about actual finite-dimensional probability laws and centered covariances. Its positive and negative conclusions concern different quantifiers: fixed-parameter mixing versus uniform contraction over parameters. The proof will retain δ_β throughout. Positive integral operators have a classical Perron–Jentzsch theory, including Birkhoff's projective treatment [1]. Our elementary power construction makes the required input explicit. Lüscher's physical transfer construction [2] additionally identifies a gauge-invariant quantum state space and a lattice Hamiltonian; those identifications are not asserted for this seven-link Markov model.

2 Haar coordinates and the three exact channels

If $V = (a, b) \in \mathbb{R} \times \mathbb{R}^3$ is a unit quaternion, then $\tau(V) = a$. Uniform measure on the unit sphere in $\mathbb{R}^4$ is Haar measure on G. Its coordinate symmetries give

$$\int a\, dV = 0, \quad \int a^2\, dV = \frac{1}{4}, \quad \int a^4\, dV = \frac{1}{8}, \qquad \int a_i a_j\, dV = \frac{\delta_{ij}}{4}. \tag{8}$$

For the fourth moment, rotational symmetry gives $\mathbb{E}a_i^4 = 3\mathbb{E}a_i^2 a_j^2$ for $i \neq j$, by rotating $(a_i + a_j)/\sqrt{2}$. Expanding $(\sum_{i=0}^3 a_i^2)^2 = 1$ then gives $4\mathbb{E}a_0^4 + 12\mathbb{E}a_0^2 a_1^2 = 1$ and the stated value.

The tilt (1) depends only on a, so

$$\mathbb{E}_{\mu_\beta} b_i = \mathbb{E}_{\mu_\beta} a b_i = 0, \qquad \mathbb{E}_{\mu_\beta} b_i b_j = \frac{1 - m_\beta}{3} \delta_{ij}. \tag{9}$$

Consequently, for each fixed $Q \in G$,

$$\int \tau(QV)\, d\mu_\beta(V) = c\tau(Q), \tag{10}$$

$$\int [\tau(QV)^2 - 1/4]\, d\mu_\beta(V) = r[\tau(Q)^2 - 1/4]. \tag{11}$$

Indeed $\tau(QV) = a_Q a - b_Q \cdot b$, and (9) evaluates both expressions. More generally, right multiplication by the tilted quaternion contracts every linear form in the four quaternion coordinates by c, and every traceless quadratic form by r. For the latter assertion, let O_V be the orthogonal matrix of right multiplication. Averaging $O_V^T T O_V$ for a symmetric matrix T gives $rT + (1 - r)\,\mathrm{tr}(T) I/4$. This follows by writing $O_V = aI + \sum b_i J_i$, using the vanishing mixed moments and $\sum_{i=1}^3 J_i^T T J_i = \mathrm{tr}(T) I - T$ for symmetric T; the identity is checked on diagonal and symmetric off-diagonal matrix units using the quaternion multiplication table.

Lemma 2.1. *For every finite β, $|c| < 1$, $|r| < 1$, and $c^2 \le m_\beta < 1$. Also $c_\beta \to 1$ as $\beta \to +\infty$.*

Proof. Cauchy–Schwarz gives $c^2 \le m_\beta$. The tilted measure has a strictly positive density relative to Haar measure, and $a^2 < 1$ on a nonempty open set, so $m_\beta < 1$. Hence $|c| < 1$ and $-1/3 \le r < 1$, implying $|r| < 1$. For $\varepsilon > 0$, the upper cap $\{a > 1 - \varepsilon/2\}$ has Haar mass $b_\varepsilon > 0$, and comparison of numerator and partition function gives, for $\beta \ge 0$,

$$\mu_\beta\{a \le 1 - \varepsilon\} \le b_\varepsilon^{-1} e^{-\beta\varepsilon/2}.$$

Thus $0 \le 1 - c_\beta \le \varepsilon + 2 b_\varepsilon^{-1} e^{-\beta\varepsilon/2}$. First let $\beta \to \infty$ and then $\varepsilon \downarrow 0$. □

Under product Haar measure, the three variables L, R, U are independent Haar variables. Each path is a product of independent Haar links, and the two paths and shared link use disjoint coordinates. Quaternion coordinate integration gives

$$\int_G \tau(PV)\tau(QV)\, dV = \frac{\tau(PQ^{-1})}{4}. \tag{12}$$

Changing variables $V = LU$ in this formula proves $\int_G x(s) y(s)\, dU = z(s)$ for fixed L, R.

Proposition 2.2. *The functions y, z, w are orthogonal in $L^2(H)$. Their squared norms are $1/4$, $1/64$ and $3/64$, respectively. Their exact kinetic evolutions are*

$$K_\beta y = c^4 y, \qquad K_\beta z = c^6 z, \qquad K_\beta w = c^6 r\, w. \tag{13}$$

Proof. For fixed L, U, write $x = \tau(LU)$. Integration in the independent right-path Haar variable R gives

$$\mathbb{E}_R y^2 = \tfrac{1}{4}, \qquad \mathbb{E}_R z^2 = \tfrac{1}{64}, \qquad \mathbb{E}_R yz = \tfrac{x}{16}.$$

Since $w = xy - z$, this also gives $\mathbb{E}_R yw = 3x/16$, $\mathbb{E}_R zw = x^2/16 - 1/64$ and $\mathbb{E}_R w^2 = x^2/8 + 1/64$. Averaging over L, U and using $\mathbb{E}x = 0$, $\mathbb{E}x^2 = 1/4$ proves all norms and cross products.

Each of the four independent link increments in the right plaquette contributes the factor c in (10); inverses have the same law, so this proves the first evolution. The six outer increments

in $\tau(LR)$ similarly give c^6. For the last evolution, first integrate the six outer increments. The product xy acquires c^6 and so does z. As a function of the remaining shared increment, $xy - z$ is a quadratic form in its quaternion coordinates with zero Haar mean. It is therefore a traceless quadratic form, contracted by r according to the calculation following (11). This shared link is integrated once, giving $c^6 r$ rather than a factor obtained by pretending the two plaquettes have independent shared links. □

The law μ_β is invariant under inversion and conjugation. Thus K_β preserves Haar measure and is selfadjoint on $L^2(H)$, by the change of variables $(s, v) \mapsto (sv, v^{-1})$ in the bounded pairing. Jensen's inequality gives its L^2 contraction; bounded functions are dense, so these statements extend to the whole space. The operator $S_{\beta,t}$ is bounded and selfadjoint. Positive integral kernels below mean pointwise positive densities, not an unproved assertion that every L^2 eigenvalue is nonnegative.

3 A ground function constructed from positive kernel powers

We give the construction because a conditional ground-state formula cannot alone establish the stationary assertions in Theorem 1.1.

Lemma 3.1. *Let (X, ν) be a nonempty probability space and let $k : X \times X \to \mathbb{R}$ be jointly measurable with $0 < m \le k(x, y) \le M < \infty$ everywhere. Then $Tf(x) = \int k(x, y) f(y)\, d\nu(y)$ has a measurable eigenfunction ψ and eigenvalue Λ satisfying*

$$m \le \Lambda \le M, \qquad m/M \le \psi(x) \le M/m, \qquad T\psi = \Lambda\psi.$$

The construction requires no supplied eigenfunction or spectral radius.

Proof. Set $p_0 = 1$, $p_{n+1} = Tp_n$ and $r_n = p_{n+1}/p_n$. Then $m^n \le p_n \le M^n$, and positivity gives $m \le r_n \le M$ by induction: $mp_n \le p_{n+1} \le Mp_n$ holds at $n = 0$ and is preserved by T. Let $\nu_n = p_n \nu / \int p_n\, d\nu$. The exact ratio relation is

$$r_{n+1}(x) = \frac{\int k(x, y) r_n(y)\, d\nu_n(y)}{\int k(x, y)\, d\nu_n(y)}.$$

The averaging density in this expression is at least $\varepsilon = m/M$. Subtracting $\varepsilon\nu_n$ leaves a positive measure of mass $1 - \varepsilon$. If $r_n \in [a_n, b_n]$, it follows that

$$r_{n+1} \in [a_{n+1}, b_{n+1}], \qquad (a_{n+1}, b_{n+1}) = \varepsilon \int r_n\, d\nu_n + (1 - \varepsilon)(a_n, b_n).$$

Starting with $[a_0, b_0] = [m, M]$, these intervals are nested and $b_n - a_n = (M - m)(1 - \varepsilon)^n$. They shrink to one common $\Lambda \in [m, M]$, so $r_n \to \Lambda$ uniformly.

Fix $x_0 \in X$ and set $q_n(x) = p_n(x)/p_n(x_0)$. The comparison $m \int p_{n-1} \le p_n(x) \le M \int p_{n-1}$ gives $m/M \le q_n \le M/m$ for $n \ge 1$, and also for $n = 0$. Since $q_{n+1}(x) = q_n(x) r_n(x)/r_n(x_0)$, we have

$$|q_{n+1}(x) - q_n(x)| \le \frac{M(M - m)}{m^2}(1 - \varepsilon)^n.$$

The geometric series proves uniform convergence to a measurable ψ with the stated bounds. The identity $Tq_n = r_n(x_0) q_{n+1}$ passes to the limit by dominated convergence, giving $T\psi = \Lambda\psi$ pointwise. The case $m = M$ is included: the kernel is constant and all subsequent ratios and normalized powers are constant. □

For (4), the actual symmetric density relative to H is

$$k_{\beta,t}(s,r) = h_t(s)h_t(r)Z_\beta^{-7}\exp\left\{\beta\sum_{e=1}^{7}\tau(s_e^{-1}r_e)\right\}. \tag{14}$$

It satisfies the everywhere bounds

$$m_{\beta,t} = \frac{e^{-|t|-7|\beta|}}{Z_\beta^7} \le k_{\beta,t} \le \frac{e^{|t|+7|\beta|}}{Z_\beta^7} = M_{\beta,t}. \tag{15}$$

Lemma 3.1 therefore proves actual ground-function existence for every real β, t. Formula (5) is a genuine probability kernel, since its density integrates to one by the eigenfunction equation. Detailed balance follows because

$$\pi(ds)P(s,dr) = \frac{\psi(s)k_{\beta,t}(s,r)\psi(r)}{\Lambda\int\psi^2\,dH}\,dH(s)dH(r)$$

is symmetric. In particular π is stationary, and P is a selfadjoint contraction on the full $L^2(\pi)$ space.

4 Symmetries, centering and the actual observable

For $g = (g_0,\dots,g_5) \in G^6$, the gauge action on an oriented edge $i \to j$ is $s_e \mapsto g_i s_e g_j^{-1}$. The words LU, $U^{-1}R$ and LR are closed words based at vertex 0, so their half traces, and hence x, y, z, w, are gauge invariant. Product Haar measure is invariant under this action. Simultaneous transformation of s, r in (14) conjugates each $s_e^{-1}r_e$ by the gauge element at its terminal vertex; τ is conjugation invariant. Thus

$$k_{\beta,t}(gs,gr) = k_{\beta,t}(s,r), \qquad S(f\circ g) = (Sf)\circ g. \tag{16}$$

Let J be the center flip $D \mapsto -D$ on the first private link of the right path. Then $J^2 = I$, H is J-invariant, and

$$x\circ J = x, \qquad (y,z,w)\circ J = -(y,z,w).$$

The operator K_β commutes with this left multiplication, since $J(s)v = J(sv)$; its two spatial weights are also unchanged. Replacing a positive ground function by $(\psi + \psi\circ J)/2$ therefore preserves its eigenvalue, lower and upper bounds, and makes it J-even. This replacement is made from now on.

Lemma 4.1. *Every uniformly positive bounded measurable eigenfunction of S is gauge invariant. In particular the chosen even ψ, the stationary measure π and its kernel P have the stated gauge covariance.*

Proof. Fix a gauge transformation g. The function $\chi = \psi\circ g$ has the same eigenvalue by (16). The ratio $F = \chi/\psi$ is bounded and satisfies $PF = F$. The ground transform can be written as normalized averaging of the same kernel k against the probability

$$d\nu_\psi = \frac{\psi\,dH}{\int\psi\,dH}: \quad Pf(s) = \frac{\int k(s,r)f(r)\,d\nu_\psi(r)}{\int k(s,r)\,d\nu_\psi(r)}. \tag{17}$$

The averaging density is at least $m_{\beta,t}/M_{\beta,t} = \varepsilon > 0$. If F lies in $[a,b]$, subtracting the common $\varepsilon\nu_\psi$ part shows that the range of PF has width at most $(1-\varepsilon)(b-a)$. Since $PF = F$, the oscillation

of F is zero, and $\chi = C\psi$ pointwise. Gauge invariance of H gives $\int \chi \, dH = \int \psi \, dH > 0$, forcing $C = 1$. The density formula for π and (16) now prove invariance of the measure and covariance of the kernel. Notice that ν_ψ in (17) is generally different from the stationary measure with density proportional to ψ^2. □

For real A, B define the concrete odd trial

$$p_t = y + tAz + tBw, \qquad F_t = h_t^{-1} p_t, \qquad f_t = F_t/\psi. \tag{18}$$

The trial and observable are bounded and measurable. The lower bound on ψ is needed for boundedness of f_t. They are gauge invariant, F_t is J-odd, and ψ is J-even. Therefore

$$\int f_t \, d\pi = \frac{\int F_t \psi \, dH}{\int \psi^2 \, dH} = 0. \tag{19}$$

The exact norm and lag relation is

$$\frac{\int f_t P f_t \, d\pi}{\int f_t^2 \, d\pi} = \frac{\langle F_t, S F_t \rangle_H}{\Lambda \|F_t\|_H^2}. \tag{20}$$

Both numerator and denominator follow by cancelling the positive ψ factors in (5). This equality uses the actual ground normalization; a pointwise row normalization of S would define a different operator and would not give this formula.

5 A positive supersolution with a cubic error

We first bound Λ above on a nonempty interval of spatial parameters. This will supply the normalization in the lower trial test. Set

$$a = c^4, \quad b = r^4, \qquad u = \frac{a}{1-a}, \quad v = \frac{b(1+a)}{2(1-a)(1-b)}, \quad \mu = \frac{1+a}{8(1-a)}. \tag{21}$$

The denominators are nonzero by Lemma 2.1. For $|q| \le 1$ let

$$P_t(q) = 1 + utq + vt^2(q^2 - 1/4), \tag{22}$$

$$C_0(A,B) = (2+e)(1 + |A| + |B|),$$

$$C = C_0(u,v) + |\mu|(|u| + |v|), \qquad Q_t = 1 + \mu t^2 + 2C|t|^3. \tag{23}$$

Lemma 5.1. *For $|t| \le 1$ and $|q| \le 1$,*

$$\left| e^{tq}(1 + Atq + Bt^2(q^2 - 1/4)) - \left[1 + (1+A)tq + t^2\{(B + A + 1/2)q^2 - B/4\}\right] \right| \le C_0(A,B)|t|^3. \tag{24}$$

If a probability-distributed Y takes values in $[-1,1]$ and has $\mathbb{E}Y = aq$, $\mathbb{E}Y^2 = 1/4 + b(q^2 - 1/4)$, then

$$|\mathbb{E}[e^{tY} P_t(Y)] - (1 + \mu t^2) P_t(q)| \le C|t|^3. \tag{25}$$

If $|u||t| + |v||t|^2 \le 1/2$, then $P_t(q) \ge 1/2$ and $\mathbb{E}[e^{tY} P_t(Y)] \le Q_t P_t(q)$.

Proof. Taylor's series gives $|e^{tq} - 1 - tq - (tq)^2/2| \le e|t|^3$ on the stated rectangle. Multiplication by $1 + Atq + Bt^2(q^2 - 1/4)$ has absolute value at most $1 + |A| + |B|$, since $|q^2 - 1/4| \le 1$. The remaining terms of total degree three and four are

$$t^3[Bq(q^2 - 1/4) + Aq^3/2] + t^4 Bq^2(q^2 - 1/4)/2,$$

bounded by $|t|^3(3|B|/2 + |A|/2)$. This proves (24) with the displayed larger constant.

Substitute $A = u, B = v$ and the two moments of Y. The definitions $a(1 + u) = u$ and $b(v + u + 1/2) = v$ give exactly

$$\mathbb{E}\left[1 + (1 + u)tY + t^2\{(v + u + 1/2)Y^2 - v/4\}\right] = P_t(q) + \mu t^2.$$

Also $|P_t(q) - 1| \le |t|(|u| + |v|)$ for $|t| \le 1$. The difference between $P_t + \mu t^2$ and $(1 + \mu t^2)P_t$ is therefore bounded by $|\mu|(|u| + |v|)|t|^3$, proving (25). The smallness condition gives $P_t \ge 1/2$, which absorbs the additive $C|t|^3$ into $2C|t|^3 P_t$ and proves the last assertion. □

Proposition 5.2. *If $|t| \le 1$ and $|u||t| + |v||t|^2 \le 1/2$, then $\Lambda \le Q_t$.*

Proof. For $q = x(s)$ the random variable $Y = x(sV)$ under the seven-link increment law has the moments in Lemma 5.1: the four increments in the first plaquette give $a = c^4$ and $b = r^4$ by (10)–(11). Inverses and conjugations preserve the central increment law, so these moment computations are valid for every fixed starting configuration. Thus $U_t(s) = h_t(s)P_t(x(s))$ satisfies $SU_t \le Q_t U_t$. It is bounded and at least $e^{-|t|/2}/2$.

For any bounded measurable R, symmetry and positivity of k give

$$\begin{aligned}\langle U_t R, SU_t R\rangle_H &\le \frac{1}{2}\iint k(s,r)U_t(s)U_t(r)[R(s)^2 + R(r)^2], dH(s)dH(r)\\ &= \int U_t(s)(SU_t)(s)R(s)^2\, dH(s) \le Q_t\|U_t R\|_H^2.\end{aligned}$$

The first inequality is $2R(s)R(r) \le R(s)^2 + R(r)^2$. Take $R = \psi/U_t$, which is bounded by the positive lower bound on U_t. The eigenfunction equation and $\|\psi\|_H > 0$ give $\Lambda \le Q_t$. Neither a formal eigenbranch nor a spectral identification of the approximate polynomial is used. □

6 The exact odd numerator and denominator

The inverse half weight in (18) cancels the two spatial weights in the kinetic numerator. Proposition 2.2 gives

$$N_t := \langle F_t, SF_t\rangle_H = \frac{c^4}{4} + \frac{t^2}{64}(c^6A^2 + 3c^6 rB^2). \tag{26}$$

There is no remainder in this identity. All exponential error lies in the denominator $D_t = \|F_t\|_H^2 = \int e^{-tx}p_t^2\, dH$.

Lemma 6.1. *For all real t, A, B, one has $D_t \ge e^{-|t|}/4 > 0$ and*

$$D_t = \int_G e^{-t\tau(V)}\mathcal{R}_t(\tau(V))\, dV, \quad \mathcal{R}_t(q) = \frac{1}{4} + \frac{t(A + 3B)q}{8} + \frac{t^2}{64}[(A - B)^2 + 8B(A + B)q^2]. \tag{27}$$

Put

$$d = \frac{A^2 + 3B^2 - 2A - 6B + 2}{64},$$
$$R_0 = \frac{(A-B)^2}{64} + \frac{B(A+B)}{32},$$
$$E = \frac{1}{4} C_0(-(A+3B)/2, B(A+B)/2) + 2|R_0|. \tag{28}$$

Then, for $|t| \le 1$,

$$|D_t - (1/4 + dt^2)| \le E|t|^3. \tag{29}$$

Proof. For fixed L, U, $x = \tau(LU)$ and $p_t = (1 + tBx)y + t(A - B)z$. The three conditional moments in the proof of Proposition 2.2 therefore give $\mathbb{E}_R p_t^2 = \mathcal{R}_t(x)$. Haar invariance in L makes LU a Haar variable, proving (27). The exact global Gram identity also gives $\int p_t^2 \, dH = 1/4 + t^2(A^2 + 3B^2)/64 \ge 1/4$; since $e^{-tx} \ge e^{-|t|}$, the positive lower bound follows.

Set $A_0 = -(A + 3B)/2$, $B_0 = B(A + B)/2$. Direct expansion shows

$$\mathcal{R}_t(q) = \tfrac{1}{4}[1 + A_0(-t)q + B_0(-t)^2(q^2 - 1/4)] + t^2 R_0.$$

Apply (24) with parameter $-t$ to the bracket. In the last term use $|e^{-tq} - 1| \le 2|t|$ for $|t|, |q| \le 1$. The total error is at most $E|t|^3$. The integrated linear part vanishes by $\int \tau = 0$; the quadratic part is dt^2 by $\int \tau^2 = 1/4$. This proves (29) with the stated constant and both signs of t. □

7 An optimized positive margin and a genuine spatial interval

Assume $c \ne 0$ and put $\alpha = c^2$. Fix the actual trial parameters

$$A = \frac{1}{1 - \alpha}, \qquad B = \frac{1}{1 - \alpha r}. \tag{30}$$

Their denominators are strictly positive by Lemma 2.1. Let d, E be (28) for this pair, and define

$$\Delta = \frac{c^6 A^2 + 3c^6 r B^2}{64} - c^4 d - \frac{c^4 \mu}{4}. \tag{31}$$

Lemma 7.1. *The margin is strictly positive. More explicitly,*

$$\Delta = \frac{\alpha^3[1 + 3r - \alpha(3 + r)]}{64(1 - \alpha^2)(1 - \alpha r)} > 0. \tag{32}$$

Proof. Insert (30) and $\mu = (1 + c^4)/(8(1 - c^4))$ into (31). Completing the two scalar squares in A, B gives

$$\Delta = \frac{\alpha^2}{64}\left[\frac{1}{1 - \alpha} + \frac{3}{1 - \alpha r} - \frac{4}{1 - \alpha^2}\right].$$

Putting the bracket over its common denominator yields (32). Now $0 < \alpha \le m_\beta < 1$ and $1 + 3r = 4m_\beta \ge 4\alpha$, while $r < 1$. Thus $1 + 3r - \alpha(3 + r) \ge \alpha(1 - r) > 0$. Both denominator factors are positive, proving the assertion. □

This margin compares the exact odd trial with an upper bound on the actual ground normalization. A calculation of the kinetic numerator alone would not establish (6). Define the nonnegative comparison constant

$$J = E + \mu(|d| + E) + 2C(1/4 + |d| + E), \tag{33}$$

and choose the explicit positive window

$$\delta_\beta = \min\left\{1, \frac{\Delta}{|c^4 J| + 1}, \frac{1}{2(|u| + |v| + 1)}\right\}. \tag{34}$$

For $0 < t < \delta_\beta$, both supersolution smallness and $c^4 J t < \Delta$ hold. From (29), for $0 \le t \le 1$,

$$D_t \le \tfrac{1}{4} + |d| + E, \qquad D_t - \tfrac{1}{4} \le t(|d| + E).$$

Expanding $Q_t D_t = D_t + \mu t^2 D_t + 2Ct^3 D_t$ and using these two bounds gives

$$Q_t D_t \le \tfrac{1}{4} + (d + \mu/4)t^2 + Jt^3. \tag{35}$$

Together with (26) and (31), this yields

$$N_t - c^4 Q_t D_t \ge t^2(\Delta - c^4 J t) > 0.$$

Proposition 5.2 gives $\Lambda \le Q_t$, while $D_t, \Lambda > 0$. Equation (20) therefore proves (6). Centering was already proved in (19), and positive variance follows from $D_t > 0$. This proves the one-step part of Theorem 1.1 on an actual nonempty interval, with no unquantified perturbative error. The window may shrink as β grows.

8 Finite-step persistence and gauge randomization

We need an iteration argument for the actual stationary operator, including a possible negative part of its spectrum.

Lemma 8.1. *Let T be a selfadjoint contraction on a real Hilbert space, and set $\mathcal{D}(F) = \|F\|^2 - \langle TF, F\rangle$. Then, for all $n \ge 0$,*

$$\|F\|^2 - n\mathcal{D}(F) \le \langle T^n F, F\rangle \le \|F\|^2. \tag{36}$$

Proof. Cauchy–Schwarz and contraction give $\mathcal{D} \ge 0$ and $\langle TZ, Z\rangle \ge -\|Z\|^2$. For $Z = F - TF$, expansion using selfadjointness gives

$$\mathcal{D}(F) - \mathcal{D}(TF) = \langle (I + T)Z, Z\rangle \ge 0.$$

Also $\|F\|^2 - \|TF\|^2 \le 2\mathcal{D}(F)$, because the right side minus the left side is $\|F - TF\|^2$. Iterating yields $\|T^k F\|^2 \ge \|F\|^2 - 2k\mathcal{D}(F)$. For $n = 2k$, selfadjointness gives $\langle T^{2k} F, F\rangle = \|T^k F\|^2$. For $n = 2k + 1$ it gives

$$\langle T^{2k+1} F, F\rangle = \|T^k F\|^2 - \mathcal{D}(T^k F) \ge \|F\|^2 - (2k + 1)\mathcal{D}(F).$$

These are the lower bounds. The upper bound follows from $\|T^n F\| \le \|F\|$ and Cauchy–Schwarz. No nonnegative-spectrum premise has entered the proof. □

Apply the lemma with $T = P_{\beta,t}$ on $L^2(\pi_{\beta,t})$ and the observable in (18). Equation (6) gives $\mathcal{D}(f) < (1 - c_\beta^4)\|f\|^2$, proving (7). Fix an integer $n \geq 0$, $q < 1$ and a cutoff B. For $n \geq 1$, Lemma 2.1 allows $\beta > B$ so large that $1 - n(1 - c_\beta^4) > q$ and $c_\beta \neq 0$. Choose $t = \delta_\beta/2$. The n-step ratio exceeds q. For $n = 0$ it is one, and the same construction supplies a nonzero centered invariant observable. This proves nonuniformity without passing to any continuum or volume limit.

Let ν be any probability measure on G^6 and define the actual randomized Markov operator

$$\overline{P}_\nu f(s) = \int_{G^6} (Pf)(gs)\, d\nu(g). \tag{37}$$

This randomizes the starting state by a gauge action before one P step. Gauge invariance of π and stationarity for P give $\pi\overline{P}_\nu = \pi$. If f is invariant, covariance (16) makes Pf invariant too; hence $\overline{P}_\nu f = Pf$. Induction gives

$$\overline{P}_\nu^{\,n} f = P^n f \qquad (n \geq 0) \tag{38}$$

for every bounded invariant f. All the covariance ratios are unchanged, including for product Haar randomization at all six vertices. The randomized operator need not itself be selfadjoint: its action on the invariant observables has already been identified with the selfadjoint P iterates. This completes Theorem 1.1.

A single open link would give a misleading alternative test. If $F : G \to \mathbb{R}$ satisfies $F(gAh^{-1}) = F(A)$ for all independent endpoint gauges, setting $A = 1, h = 1$ gives $F(g) = F(1)$, so F is constant. The half trace of an open link is not invariant: its values at 1 and -1 are 1 and -1. The observables above use actual closed words and do not rely on an open-link mode being physical.

9 Mixing at each fixed coupling

The absence of a parameter-uniform fixed-step factor does not prevent mixing at a fixed pair of parameters.

Proposition 9.1. *For every fixed β, t and bounded measurable f with $\int f\, d\pi_{\beta,t} = 0$ and $|f| \leq B$,*

$$|P_{\beta,t}^n f(s)| \leq 2B(1 - \varepsilon_{\beta,t})^n, \qquad \varepsilon_{\beta,t} = e^{-2|t|-14|\beta|} > 0. \tag{39}$$

Proof. Use the probability ν_ψ in (17). The normalized row density of $k_{\beta,t}$ is at least $m_{\beta,t}/M_{\beta,t} = \varepsilon_{\beta,t}$ by (15); all seven Wilson partitions cancel. Thus P is the sum of a common reset $\varepsilon_{\beta,t}\nu_\psi$ and a positive residual kernel of mass $1 - \varepsilon_{\beta,t}$. It follows that

$$\operatorname{osc}(Pf) \leq (1 - \varepsilon_{\beta,t}) \operatorname{osc}(f), \quad \operatorname{osc}(f) = \sup f - \inf f.$$

Iteration bounds the oscillation by $2B(1 - \varepsilon_{\beta,t})^n$. Stationarity gives $\int P^n f\, d\pi = 0$, so zero lies between its infimum and supremum. Its absolute value is bounded by its oscillation, proving (39). If $\varepsilon = 1$, the kernel is a reset; the estimate for $n \geq 1$ is direct and the $n = 0$ bound is immediate. □

This is a bounded-observable mixing result for the actual stationary finite model. Its reset constant degenerates as β grows. The slow-mode theorem proves failure of a uniform fixed-step factor, rather than merely pointing to this weak upper estimate.

10 Why the second spatial plaquette changes the argument

To weight both plaquettes equally, replace h_t by

$$\widetilde{h}_t(s) = e^{t(x(s)+y(s))/2}, \qquad \widetilde{S}_{\beta,t} = M_{\widetilde{h}_t} K_\beta M_{\widetilde{h}_t}.$$

The old center flip sends $x + y$ to $x - y$, so it no longer preserves the weight at $t \neq 0$. At the flat configuration all links are 1; the two values of $\widetilde{h}_t$ before and after the flip are e^t and 1. This is a pointwise failure of a required symmetry, which persists for every nonzero t.

The effect on centering is realized at the genuine reset coupling $\beta = 0$. Here μ_0 is Haar and $K_0 f = \int f \, dH$, so

$$\widetilde{S}_{0,t} f = \widetilde{h}_t \int \widetilde{h}_t f \, dH, \quad \widetilde{\psi}_t = \widetilde{h}_t, \quad \widetilde{\Lambda}_t = \int e^{t(x+y)} \, dH, \quad d\widetilde{\pi}_t = \frac{e^{t(x+y)} \, dH}{\int e^{t(x+y)} \, dH}.$$

The actual ground-normalized Markov kernel is the constant reset $\widetilde{P}_t f = \int f \, d\widetilde{\pi}_t$. Bounded differentiation under the integral gives

$$\frac{d}{dt} \int y \, d\widetilde{\pi}_t \Big|_{t=0} = \int y(x+y) \, dH - \int y \, dH \int (x+y) \, dH = \frac{1}{4}. \tag{40}$$

All first moments vanish, $\int xy \, dH = 0$ by integrating the independent left Haar path, and $\int y^2 \, dH = 1/4$. The mean at zero is zero; (40) proves positive mean for arbitrarily small positive t. Under these actual stationary laws,

$$\int y \widetilde{P}_t y \, d\widetilde{\pi}_t = \left(\int y \, d\widetilde{\pi}_t \right)^2 > 0,$$

whereas every correctly centered integrable observable has exactly zero one-step lag under the reset kernel. An uncentered positive lag therefore does not obstruct mixing.

This example does not rule out a corrected two-weight trial. It shows why the one-weight orthogonality and its covariance conclusion cannot simply be transplanted. In particular, the small independent t interval in Theorem 1.1 gives no statement along a homogeneous relation between spatial and temporal couplings.

11 Provenance, classical context and remaining boundaries

Every formal source used here is frozen at the verified commit

`bab688bb6140cd6316c80680095a914ea0d346db`.

The accompanying theorem map records exact declarations, unabridged headers, file hashes and baseline links. The original full-build and transitive-axiom receipt, with its evidence hashes, accompanies the draft. The publication editor has not changed a mathematical source or run a redundant research build.

The principal sources are the shared-plaquette Haar projection and channel modules; seven-link transfer, variational, odd-trial and spatial separation; positive kernel powers; realized vacuum and original ground transform; full stationary L^2 transfer; iteration, mixing and gauge averaging; and the homogeneous-plaquette obstruction. In particular, the earlier conditional ground-data contract is discharged by the proved positive-power construction. It is not used as an existence

axiom. Quaternion coordinate calculations, the factored optimized margin, explicit smallness window and even/odd iteration reorganize the verified arguments into human proofs; they are not represented as new Lean declarations. The same exact one-weight model, parameter dependence, centering and boundedness hypotheses are retained.

Positive-kernel existence, minorization, ground transforms and character methods are classical. Birkhoff's primary introduction and uniform-positivity framework [1] provide relevant antecedents; full theorem-by-theorem historical comparison remains incomplete. Lüscher's primary construction [2] includes the physical gauge-invariant Hilbert space and transfer matrix of lattice gauge theory. Here the finite Markov covariance is not identified with a physical Schwinger function or energy spectrum. Priority for the exact optimized seven-link comparison remains unresolved after a bounded primary-literature review.

There is no volume or lattice-spacing parameter in the theorem. No uniform homogeneous coupling regime, reflection-positive continuum construction, spectral-to-physical distance identification or four-dimensional mass gap is supplied. Failure of a parameter-uniform number of these finite Markov steps does not refute a physical mass gap. The draft exposes finite-transfer mechanisms and exact hypotheses for their reuse. The active research campaign and its unresolved continuum objectives remain separate.

References

[1] G. Birkhoff, *Extensions of Jentzsch's theorem*, Transactions of the American Mathematical Society **85** (1957), 219–227. doi:10.1090/S0002-9947-1957-0087058-6.

[2] M. Lüscher, *Construction of a selfadjoint, strictly positive transfer matrix for Euclidean lattice gauge theories*, Communications in Mathematical Physics **54** (1977), 283–292. doi:10.1007/BF01614090. Primary DESY preprint.